Nations at a Glance

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Applied Data Visualization and Analysis DSIM-604

April 23rd, 2026

1. Introduction

The global country dataset centers around all the countries around the globe and how they fair individually compared to each other. The topic centers around how well countries are performing from several different analytical perspectives including wealth, health, land, environmental concerns, military personnel and much more. In a chaotic post-pandemic world filled with political turmoil and military aggression, a deep dive into the overall state of the affected countries as well as the state of the global economy offer intrigue and key insight into how the world is reacting. Nothing is more important than the world that we live in, and considering every country has its part to play in the dependent relationship that is global economics, this topic is of utmost importance.

2. Global Country Information Dataset (2023)

The global country information dataset is a world-data dataset containing 35 variables spanning every legally recognized country on the globe. It is a csv text file extracted from an excel format containing just over 49 kilobytes of data spanning dozens of quantitative and qualitative variables. Key quantitative variables include land area, birth rate, life expectancy and GDP while key qualitative variables include capitals/cities, languages and currency use. The dataset does include several null categories and needs data cleansing but holds a 10/10 usability rating while being easily transferrable as a csv file. The overall variety of demographic and economic factors as well as unique listings involving environmental numbers and education variables make it a great dataset to create and analyze from.

Qualitative Data

Field Name	Data Type	Description	Possible Values
Country	**Text/Character**	**Name of Country**	**France, Spain, Iran**
Abbreviation	**Text/Character**	**Abbreviated Name of Country**	**US, FR, ES**

Capitol/Major City	**Text/Character**	**Capitol of Countries**	**Kabul, Lisbon, Baku**
Language	**Text/Character**	**Language Spoken**	**English, French, Urdu**
Largest City	**Text/Character**	**Largest City in Country by Landmass**	**Washington D.C., Vienna, Brussels**
Currency Code	**Text/Character**	**Abbreviation for Currency used by country**	**USD, EUR, CNY**
Call Code	**Numbered Text**	**Call Code of Country for Identification**	**1, 32, 229**

Quantitative Data

Agricultural Land %	**Percentage**	**% of farmland in country**	**0.581, 0.431, 0.127**
Armed Forces Size	**Whole Number**	**Number of soldiers in Military**	**0, 2,695,000, 800**
Birth Rate	**Decimal**	**Children Born per minute**	**31.88, 14.88, 10.9**
Co2 Emissions	**Whole Number**	**Total amount of Co2 gas omitted**	**97,814, 202, 3,282**
CPI	**Decimal**	**Consumer Price Index**	**125.08, 140.95, 103.62**
CPI Change (%)	**Percentage**	**Consumer Price Index Change in %**	**0.029, 0.035, -0.043**
Density (P/Km2)	**Whole Number**	**Population per Km**	**153, 46, 467**
Fertility Rate	**Decimal**	**How many children birthed per woman**	**1.69, 1.81, 4.21**

Forested Area (%)	**Percentage**	**% of land covered by forest**	**0.224, 0.557. 0.589**
Gasoline Price	**Decimal**	**Price of gasoline**	**1.56, 0.97, 3.99**
GDP	**Whole Number**	**Gross Domestic Product Annual**	**5,209,000,000, 63,080,457,023**
Gross Primary Education Enrollment (%)	**Percentage**	**Amount of pop enrolled in primary education**	**0.34, 0.78, 0.27**
Gross Tertiary Education Enrollment (%)	**Percentage**	**Amount of pop enrolled in higher level education**	**0.22, 0.08, 0.17**
Infant Mortality	**Decimal**	**Newborns deceased per 1,000 births**	**19.2, 8.3, 6.1**
Land Area (Km2)	**Whole Number**	**Amount of land in Km2**	**765, 86,600, 13,880**
Latitude	**Decimal**	**Horizontal Geographic Coordinates**	**47.577, 50.558, 90.356**
Life Expectancy	**Decimal**	**Avg Length of life**	**72.9, 73.8, 72.3**
Longitude	**Decimal**	**Vertical Geographic Coordinates**	**40.143, 57.665, 35.643**
Maternal Mortality Ratio	**Whole Number**	**Number of female deaths per 100,000 births**	**26, 70, 173**
Minimum Wage	**Decimal**	**Legally lowest permitted hourly wage**	**5.25, 0.51, 13.59**

Out of Pocket Health Expenditure	**Percentage**	**Percent of total payment out of patient pocket**	**0.784, 0.569, 0.364**
Physicians Per Thousand	**Decimal**	**Number of doctors per 1,000 people**	**1.20, 1.72, 3.33**
Population	**Whole Number**	**Number of total people**	**2,854,191, 77,142, 38, 041, 754**
Labor Force Participation (%)	**Percentage**	**% of population with jobs in labor force**	**0.489, 0.557, 0.412**
Tax Revenue	**Percentage**	**Tax to GDP %**	**0.088, 0.356, 0.147**
Total Tax Rate	**Percentage**	**Tax %**	**0.334, 0.533, 0.554**
Unemployment Rate	**Percentage**	**% of pop without jobs**	**0.1033, 0.0459, 0.0559**
Urban Pop	**Whole Number**	**Amount of pop in big cities**	**323,784, 89,431, 60,987,417**

3. Strategic Questions

Much of my work and visualization/analysis will center around the world's 'superpowers' and first world countries pertaining to how they compare in critical categories. This isn't to say that the other second and third world countries will be ignored, but with most of the prime data being heavily skewed towards first world countries, the most accurate pooled information will come from them.

Key metrics to identify these superpowers will revolve around the annual sums of GDP as well as land and population numbers relative to military strength. Beyond just wealth and power, my visualizations and analysis will delve into how these countries perform in categories involving environmental concerns while also looking into how these countries are fairing in terms of population management. Analysis will give insight into healthcare revolving around the life expectancy and birth rates as well as how national wealth is being invested into these key areas.

For the first round of chart analysis specifically, the most important topic will be firstly identifying the first world countries for the foundation of our dashboard research. This will be done by identifying the top 20 countries in terms of annual GDP which will identify primary powerhouses like the US and China, while also identifying the fringe elite wealth countries in Europe, Asia and the Middle East. Once these nations are identified, my initial analysis will venture into surface level visualization of how these countries stand in terms of environmental preservation as well as inflation risk and wealth management.

My initial analysis will also pivot from its base analysis to answer a couple of intriguing global questions including what languages are dominating the world and how the wealth of a nation impacts its birth rates. Initial charts and visualizations answering my strategic questions will include bar charts, scatter plots, world maps and a highlight table for languages.

These are my eight strategic questions for my initial research:

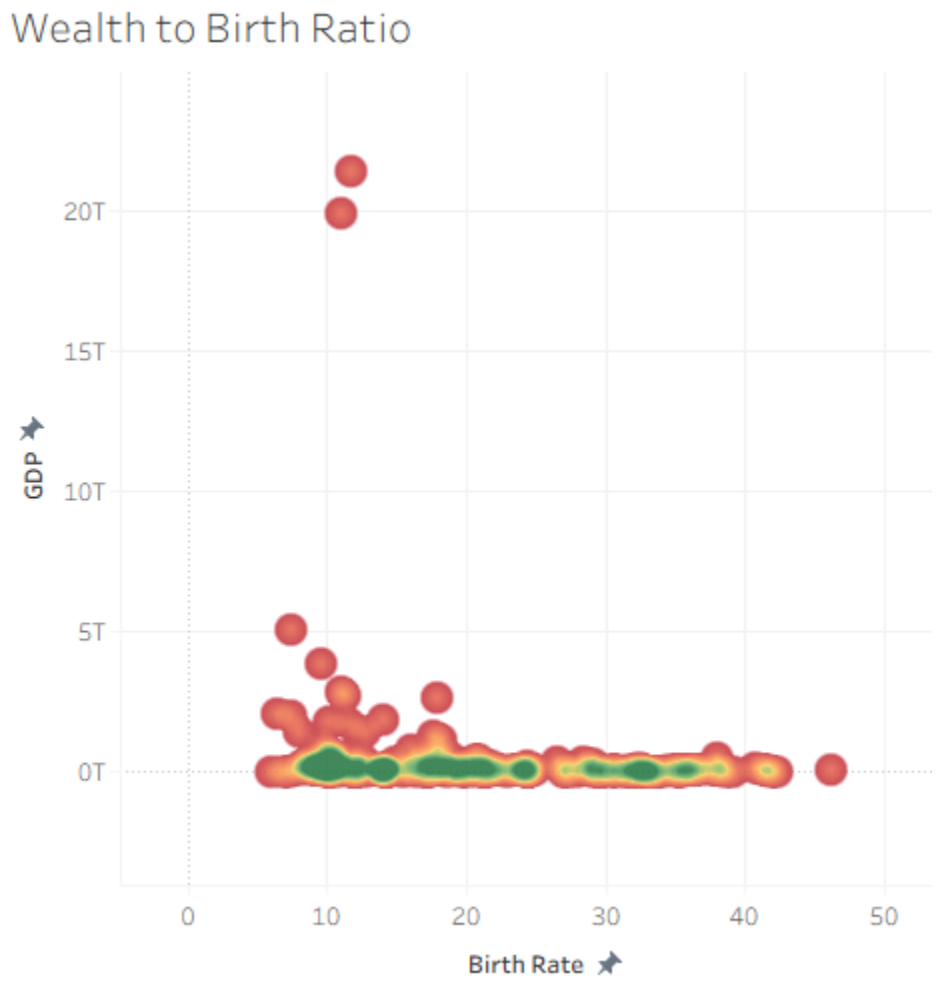
1. **Does the wealth of a nation impact the rate at which it populates?**
2. **Which countries are struggling with inflation rises post-pandemic?**
3. **Which countries are actively dominating economically in terms of GDP, and which countries are a surprisingly doing well?**
4. **Of the world's biggest economic powerhouses, which countries remain strong in their environmental initiatives?**
5. **What languages tend to be spoken the most throughout the world?**
6. **Of the world's most successful countries, how do they fair in terms of carbon emissions and which countries produce excessive amounts of CO2?**
7. **Which countries tend to produce the most physicians, and are wealthier countries more likely to produce physicians than second and third world countries?**
8. **Which countries produce the most GDP per Capita and which regions tend to hold more of these countries?**
9. **What are the primary indicators of wealth and what do they say about the leading countries in the world?**
10. **Of the top annual GDP producers worldwide, which ones are efficient and which ones rely heavily upon total population and output?**

11. **What is the directn relationship between GDP and CO2 emissions regardless of population, and does the population of a country indicate where countries fall regarding these two variables?**

4. Chart Analysis

C1.) Does the wealth of a nation impact the rate at which it populates?

The wealth to birth ratio chart below shows that the countries with the highest GDPs in the world tend to have lower birth rates and less children. Countries with average to below-average GDPs are harder to predict, likely signaling to deeper variable factors such as existing population and land area available for the people to live.

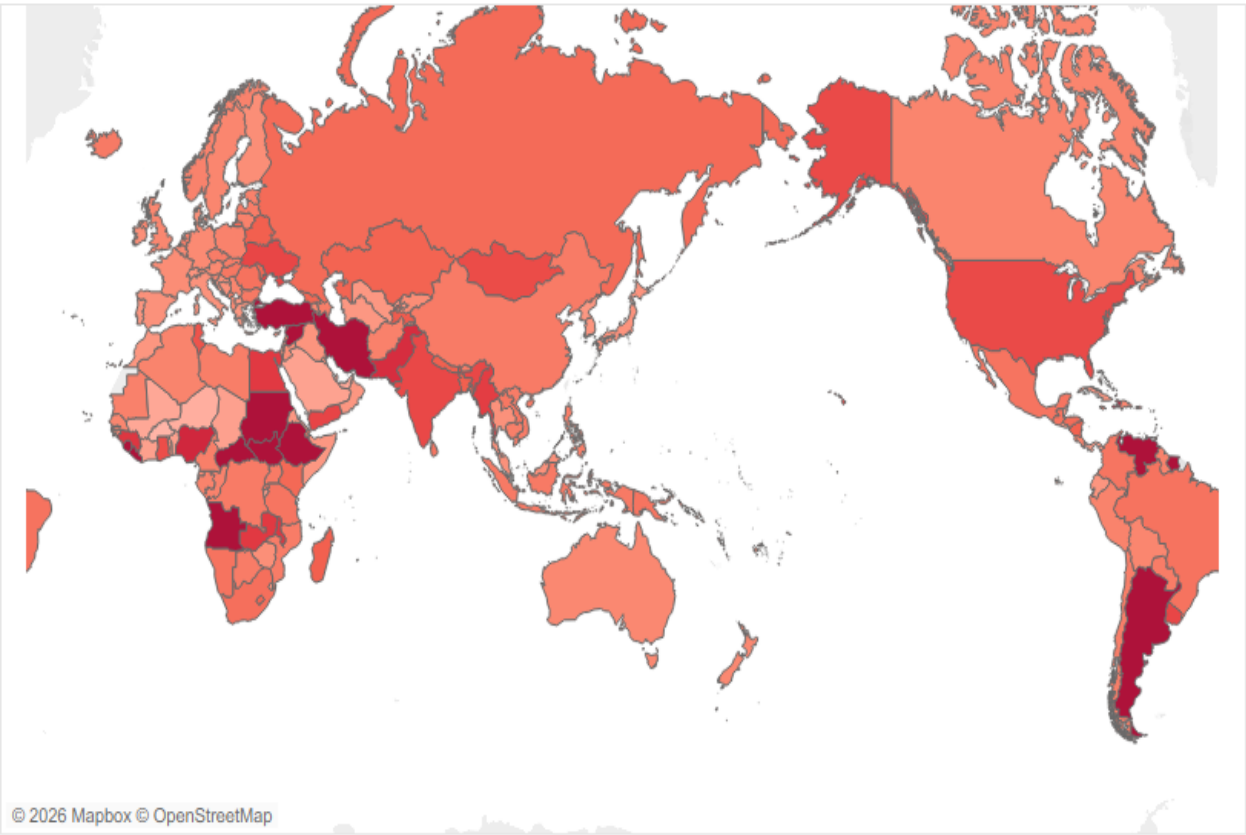


Sum of Birth Rate vs. sum of GDP. Details are shown for Country.

C2.) Which countries are struggling with inflation rises post-pandemic?

When analyzing the world map below, dark shades of red show that central African nations as well as Middle Eastern nations Turkey, Iran and Syria struggle immensely with inflation displaying abnormal jumps in Consumer Price Index (CPI) relating to their peers. South American countries Venezuela and Argentina are also struggling while the global powerhouses Russia and the USA show signs of trouble as well. With war in the middle east, political struggles in the USA and Venezuela and debt/currency problems in Africa, the map reflects seemingly accurate information.

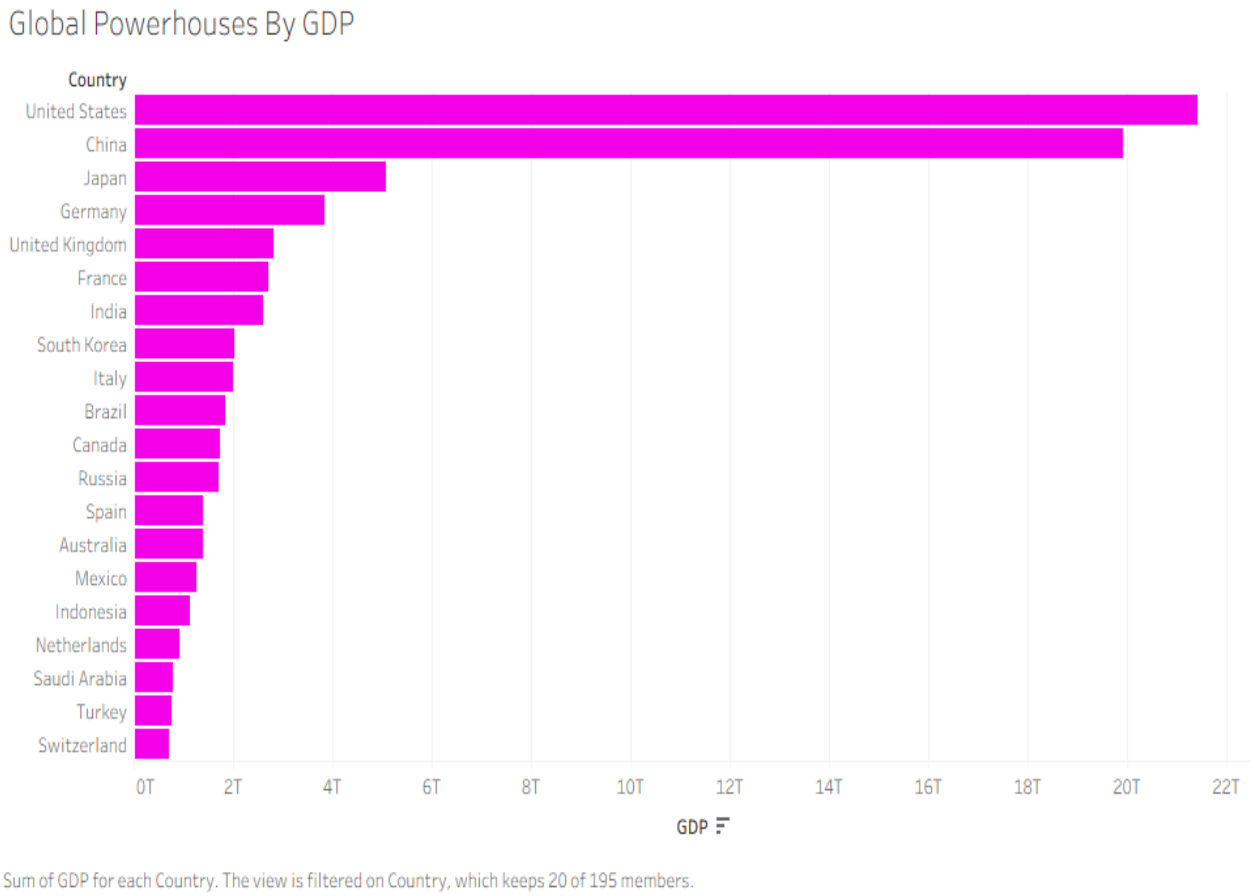
Inflation Risk



Map based on Longitude (generated) and Latitude (generated). Color shows sum of CPI Change (%). Details are shown for Country.

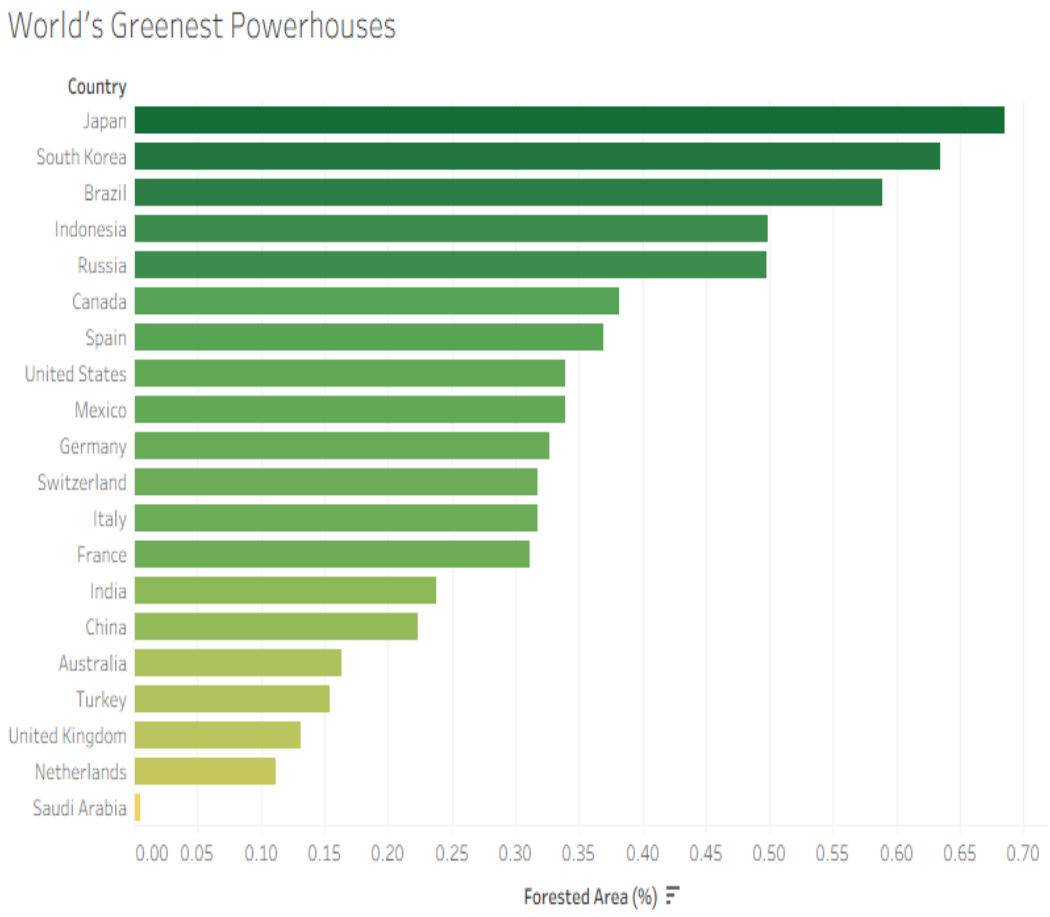
C3.) Which countries are actively dominating economically in terms of GDP, and which countries are a surprisingly doing well?

As expected, the United States (21.4T) and China (19.9T) continue to dominate well above the pack while Germany remains the backbone of European industry and engineering. India was a big surprise, having the 7th largest GDP in the world. A homegrown consumer base of over $1.4 billion and increased outsourcing work from overseas continue to feed the rise of India. Other European powers France and the United Kingdon also remain in the top 10, with surprising middle eastern nations Saudi Arabia and Turkey rounding out the top 20 GDP nations in the world.



C4.) Of the world's biggest economic powerhouses, which countries remain strong in their environmental initiatives?

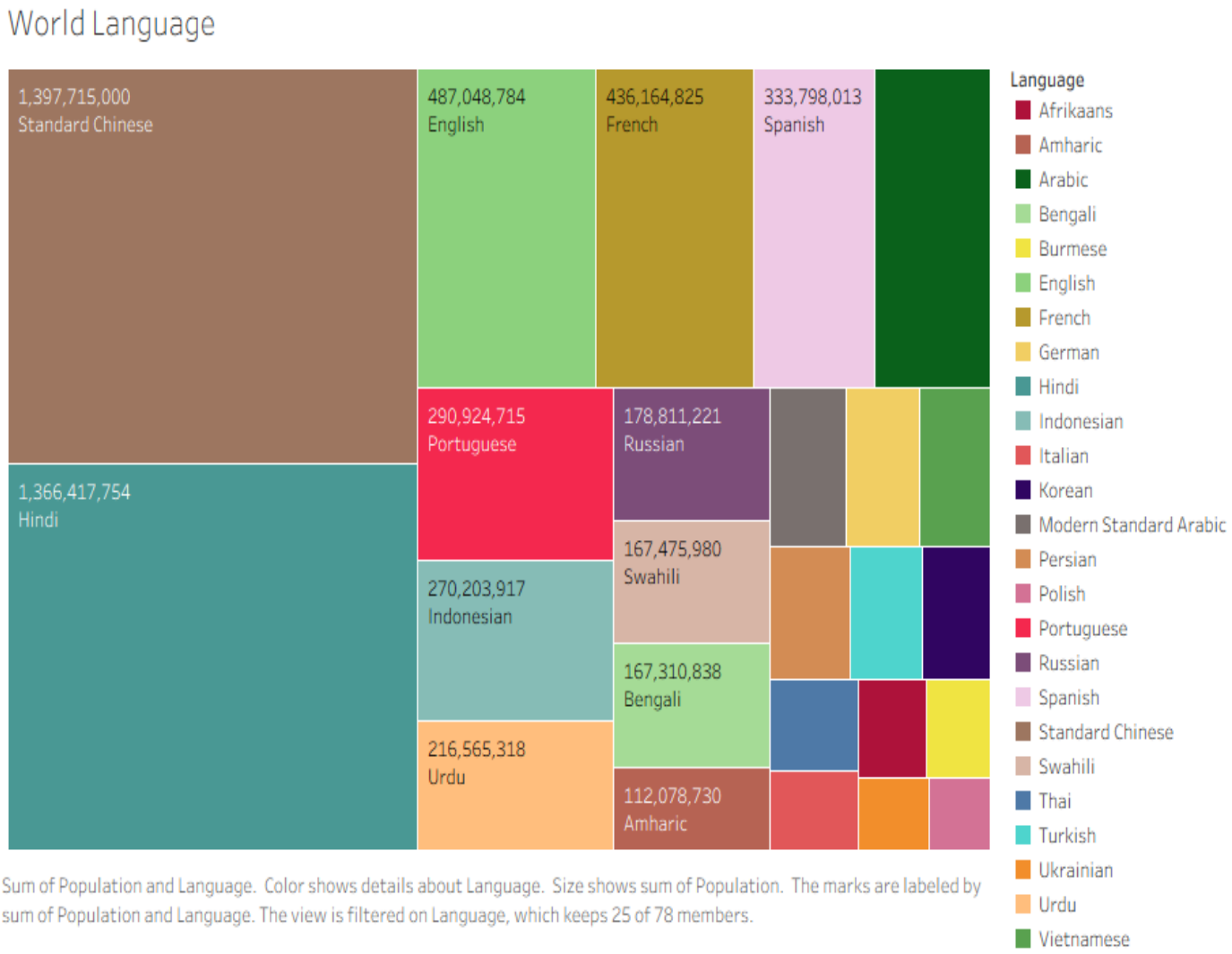
According to the chart below, the East Asian powerhouses have been the crème of the crop when it comes to keeping their environments healthy with forest landscape. Measuring the amount of land covered by forest, Japan (68.5%), South Korea (63.4%) and Indonesia (49.9%, Southeast Asia) lead the pack followed by Russia (49.8%) and Canada (38.2%). Surprisingly, the United States is above the midline of the bar chart, ranking top 8 of the powerhouses with over a third of the land being forest despite being a huge manufacturer. Brazil (58.9%) is strong also sitting at third while the Netherlands and Turkey are struggling at well below 20% of forest area.



Sum of Forested Area (%) for each Country. Color shows sum of Forested Area (%). The view is filtered on Country, which keeps 20 of 195 members.

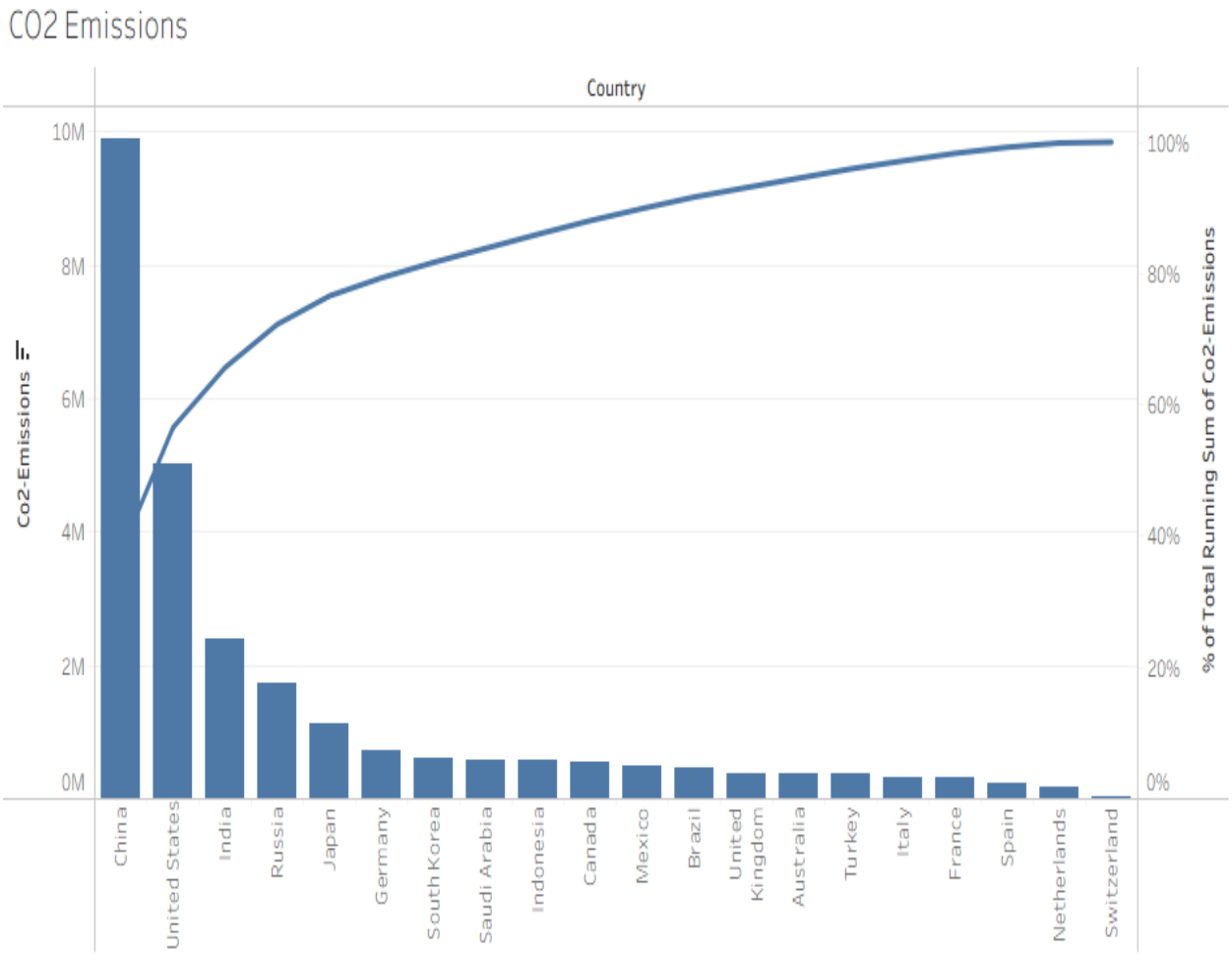
C5.) What languages tend to be spoken the most throughout the world?

Analyzing the world language chart, standard Chinese and Hindi dominate the language sphere in the world, more than doubling the amount of English speakers in the world. This should not come as a surprise as China and India have the largest populations in the world by a wide margin. English and French sit comfortably behind just under half a billion speakers each, with Spanish and Arabic rounding out the top 5 at over 300 million speakers each. Other notable languages spoken worldwide at over 100 million people include Portuguese, Indonesian, Urdu, Russian, Swahili, Bengali and Amharic.



C6.) Of the world's most successful countries, how do they fair in terms of carbon emissions and which countries produce excessive amounts of CO2?

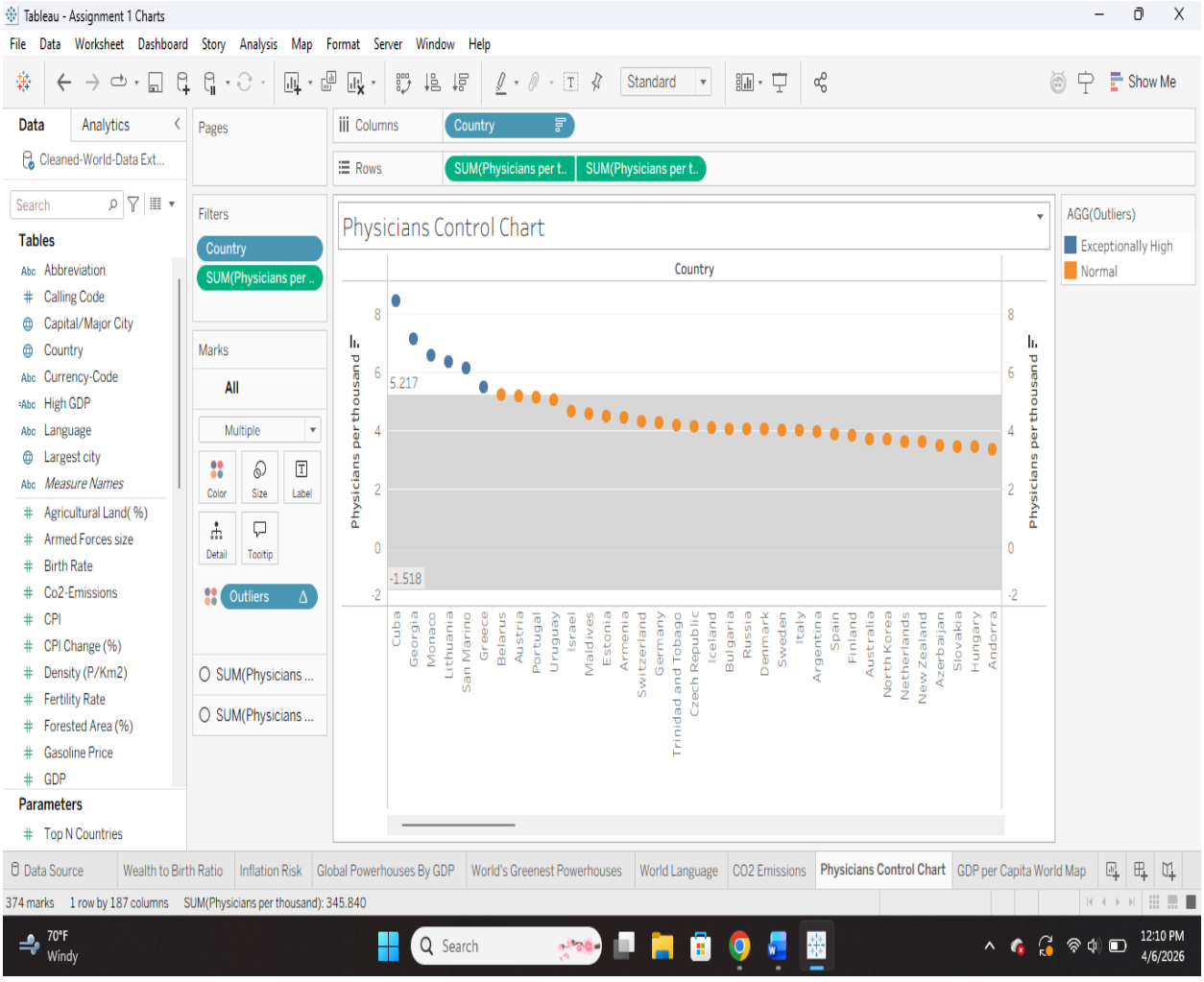
Looking at the CO2 emission pareto chart below, many of the world's powerhouse countries tend to sit in the 200,000-500,000-range making up less than 10% of the overall carbon emissions. When looking at the outliers, China is far and wide the worst country in terms of pollution, accounting for nearly 10 million emissions that stands at over a third of the total worldwide. Behind them are the United States at 5 million accounting for 20 %, with other large countries such as India, Japan and Russia accounting for over a million emissions. Together, these 5 nations account for over 76% of the total emissions. Factors leading to this large gap are likely due to the industrialization and large-scale militaries of superpower countries like the US, China and Russia.



The trends of sum of Co2-Emissions and % of Total Running Sum of Co2-Emissions for Country. The view is filtered on Country and sum of Co2-Emissions. The Country filter keeps 20 of 195 members. The sum of Co2-Emissions filter keeps non-Null values only.

C7.) Which countries tend to produce the most physicians, and are wealthier countries more likely to produce physicians than second and third world countries?

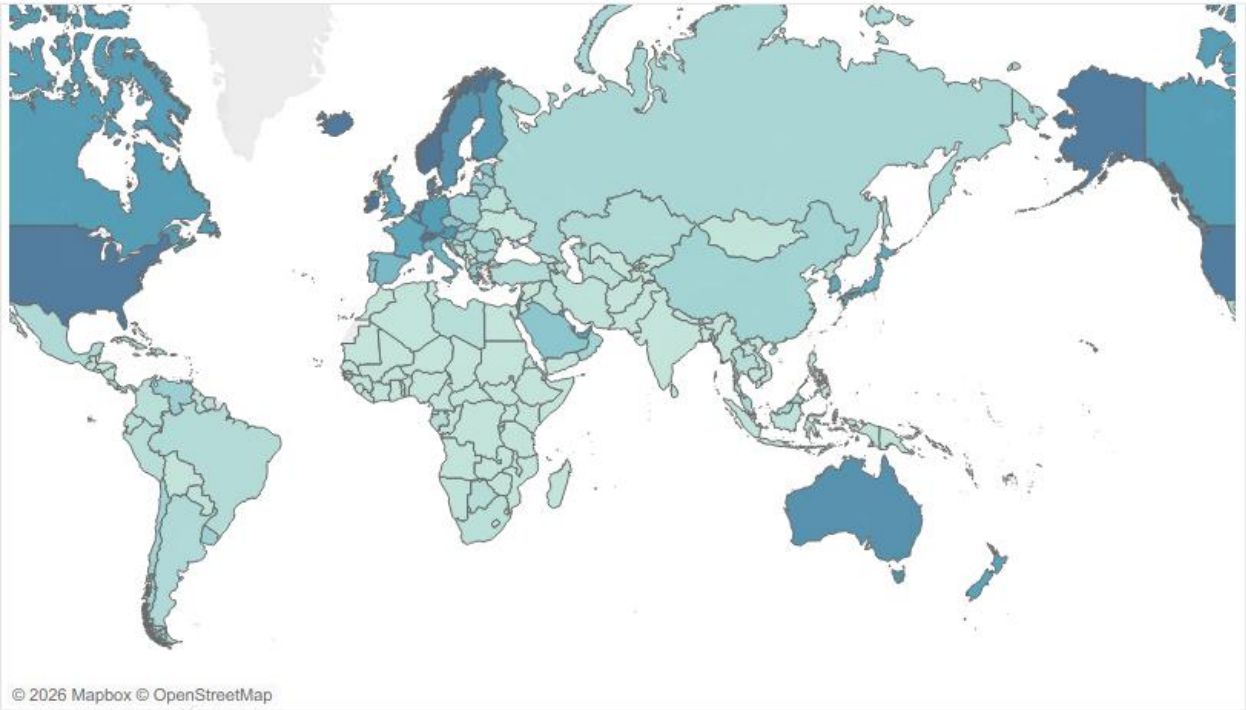
According to the control chart highlighting the count of physicians per thousand people, the countries of Cuba, Monaco, Greece, Lithuania, Georgia and San Marino are outliers producing more physicians than the statistical average. This is actually quite intriguing, as wealthy and powerful countries such as the US, China and the UK actually produce pedestrian physician numbers. When looking into why these specific countries perform above expectations, it is likely due to social status, specialized hubs and former soviet union classification. Georgia and Lithuania continue their objective as former Soviet Union specialist, while Cuba produces physicians as exports and Greece simply encourages the profession going back thousands of years. Monaco is the only wealthy country of the six to stand as an outlier.



C8.) Which countries produce the most GDP per Capita and which regions tend to hold more of these countries?

As expected, the United States is a global leader in GDP per Capita, with its neighbor Canada being a strong candidate as well. Northern European countries are doing very well too, with countries like Norway, Finland, Sweden and Iceland being world leaders in GDP per Capita. Surprisingly, two superpowers, China and Russia are rather pedestrian in GDP per Capita and only represent high GDPs due to high populations. Other surprising outliers include both Australia and New Zealand in the Pacific.

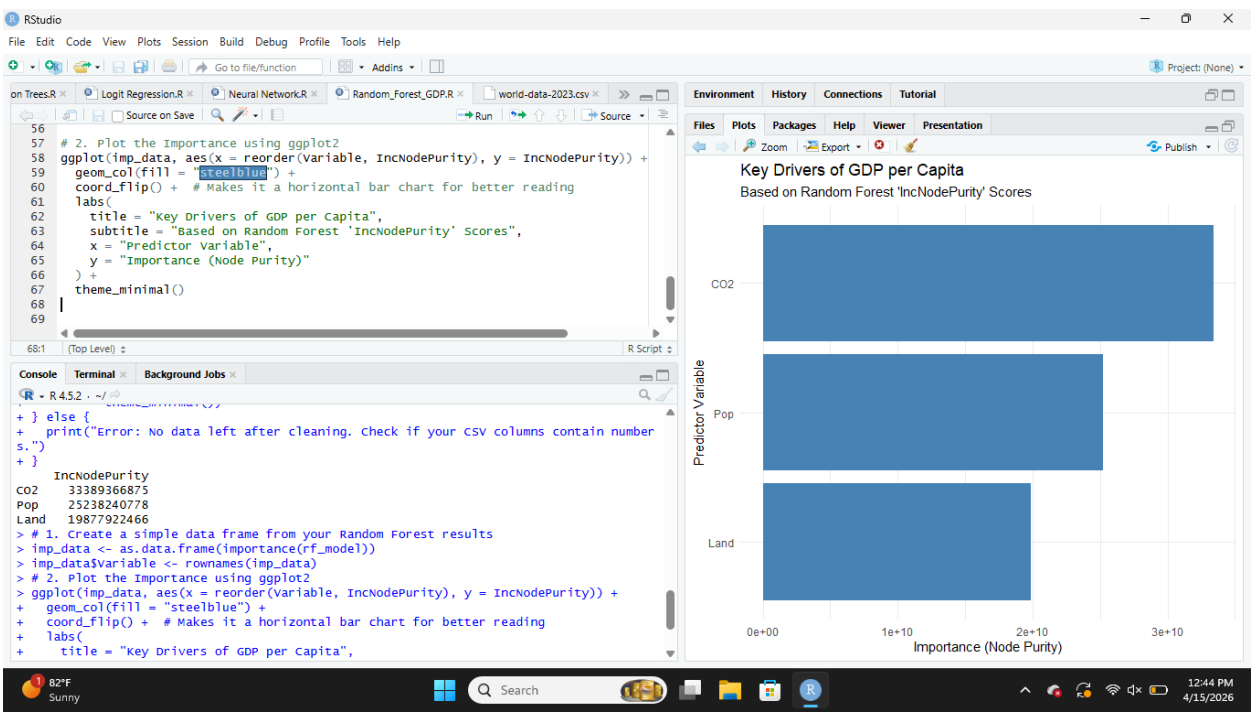
GDP per Capita World Map



Map based on Longitude (generated) and Latitude (generated). Color shows GDP per Capita. Details are shown for Country. The view is filtered on GDP per Capita, which keeps non-Null values only.

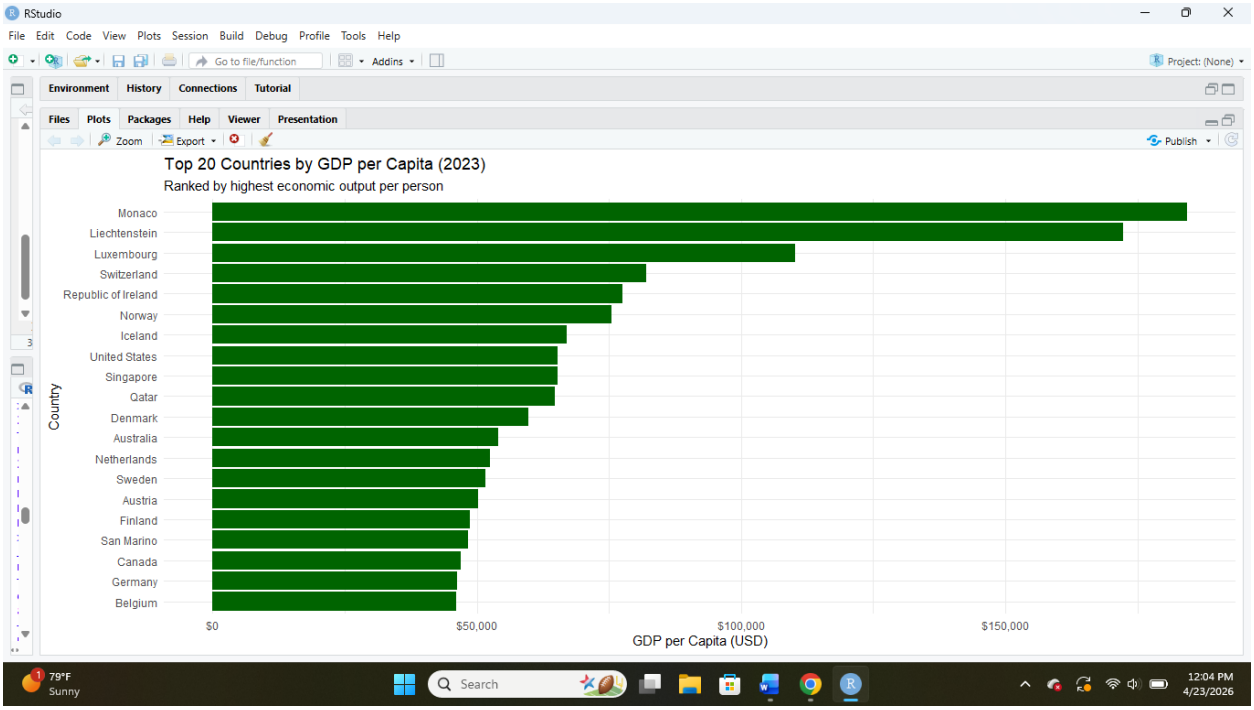
C9.) What are the primary indicators of wealth and what do they say about the leading countries in the world?

While running a random forest machine learning model, it was confirmed that carbon dioxide emissions were the leading indicator of national wealth. Population was the next leading indicator, with land being the least impactful of the big three. These results, displayed by the bar chart to the right, show that the most successful countries in the world are doing so with factory production and fossil fuel emissions driving their main exports and income. Nations like the US, China, Russia and Germany are large industrial players and consume wealth through the vast amounts generated by export and trade of the product from these emissions.



C10.) Of the top annual GDP producers worldwide, which ones are efficient and which ones rely heavily upon total population and output?

According to the descending bar chart below, many small wealthy nations such as Monaco, Luxemborg, Switzerland and Ireland are extremely efficient while large countries like China and India rely solely upon size and mass production not even cracking the top 20 GDP per Capita rankings. The US is an outlier being a large nation population wise while still maintaining the 8th highest GDP per Capita in the world. Regionally, Western Europe and North America appear to be efficient with wealth production by population while the Asian and Middle Eastern/African regions struggle to maintain high GDP per Capita statistics.

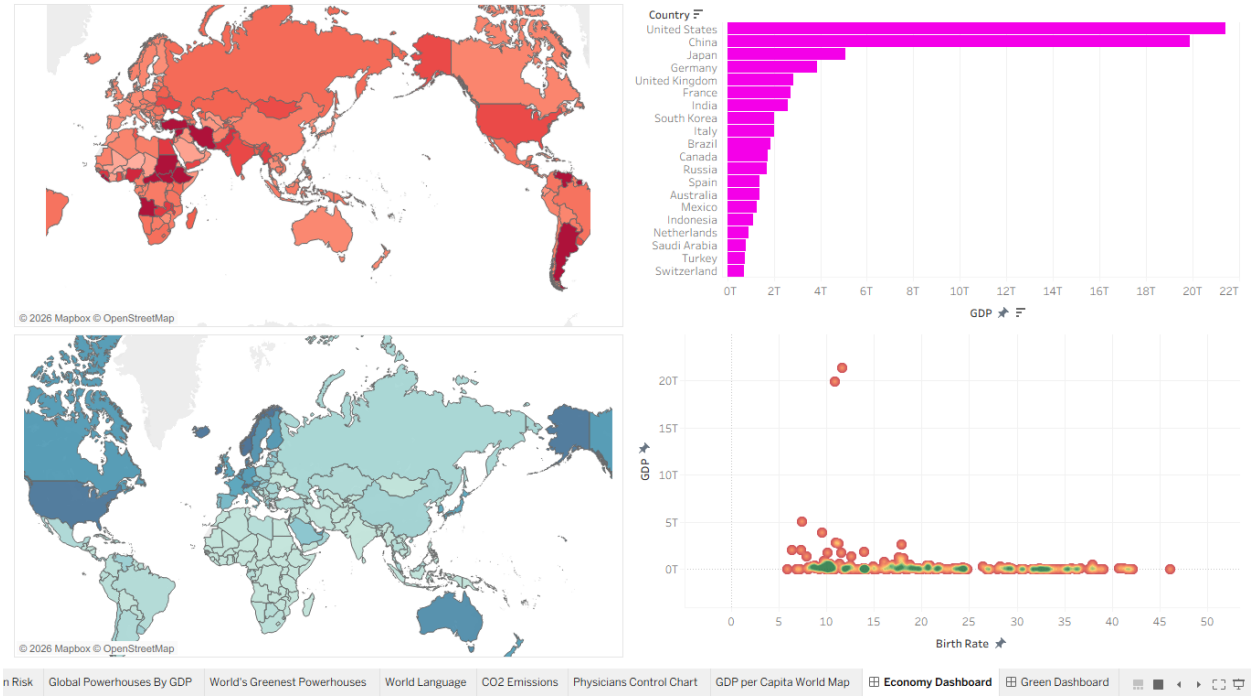


C11.) What is the direct relationship between GDP and CO2 emissions regardless of population, and does the population of a country indicate where countries fall regarding these two variables?

As shown in the interactive scatterplot below, there is a direct correlation between GDP and carbon dioxide emissions regardless of poulation size. By setting GDP and CO2 emissions to capita, the population variable was rendered non effective. Being so, wealth and carbon dioxide show a clear direct relationship, meaning where there is money made there are harmful gasses produced. Regarding the size of countries impacting their GDP to CO2 emission ratios, the result suggests there is no such correlation. Large, small and moderate countries as shown by the size of the bubble occur at random on the plot.

5. Dashboards

Economy Dashboard



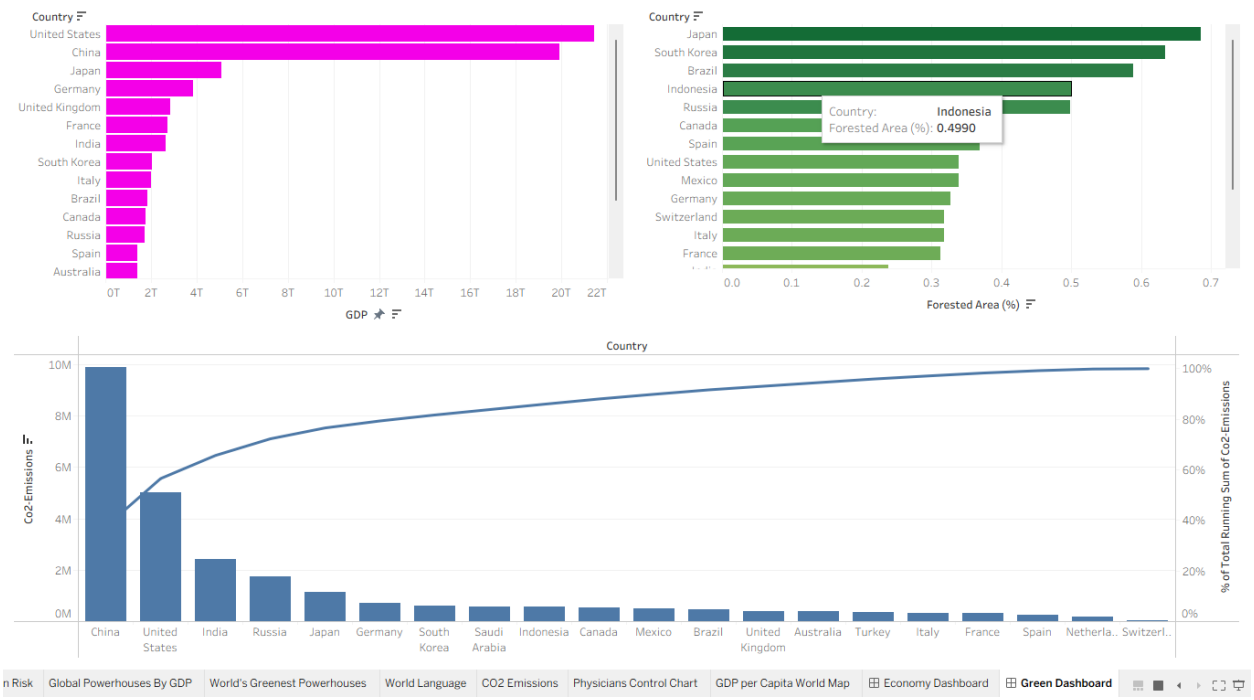
The economy dashboard above highlights not only the top GDP producing countries in the world, but also a worldwide view of GDP per capita and inflation metrics. The upper-right bar chart represents a descending order of GDP kings globally with the US and China dominating the field followed by respectable margins from Japan, Germany and the UK.

The vibrant world maps on the left showcase a global view of both inflation (top) and GDP per capita (bottom) in each county relative to its neighbors. In the red inflation visual, there are signs of clear inflation emergencies in both central Africa and South America, as well as growing concerns in the Middle East. Below, the blue visual show the clear advantage the US and its northern ally Canada have over Eastern hemisphere. Much of the heavy blue concentration

leans toward the North American continent, however western and especially northern Europe are also doing extremely well. The scandinavian nations and powerhouse european countries like Germany and the UK are dominating the East in spite of Asias massive population and production.

Lastly, a one off density plot breakdown of wealth-to-birth ratios show a relatively strong inverse relationship between the wealth of a country and its birth rates. The wealthier countries like the US, China and Japan tend to have much lower birth rates while poorer pack nations have uncontollable birth rates well into the 40s and 50s. Collectively, this economy dashboard is a detailed summarization of how wealth is being spread globally as well as what economic challenges certain countries face, particularly in the poorer eastern countries.

Green Dashboard



The environmental dashboard above, known as the “Green Dashboard”, is a summarized visual of how the top countries in the world fair from an environmental standpoint. Sure, many

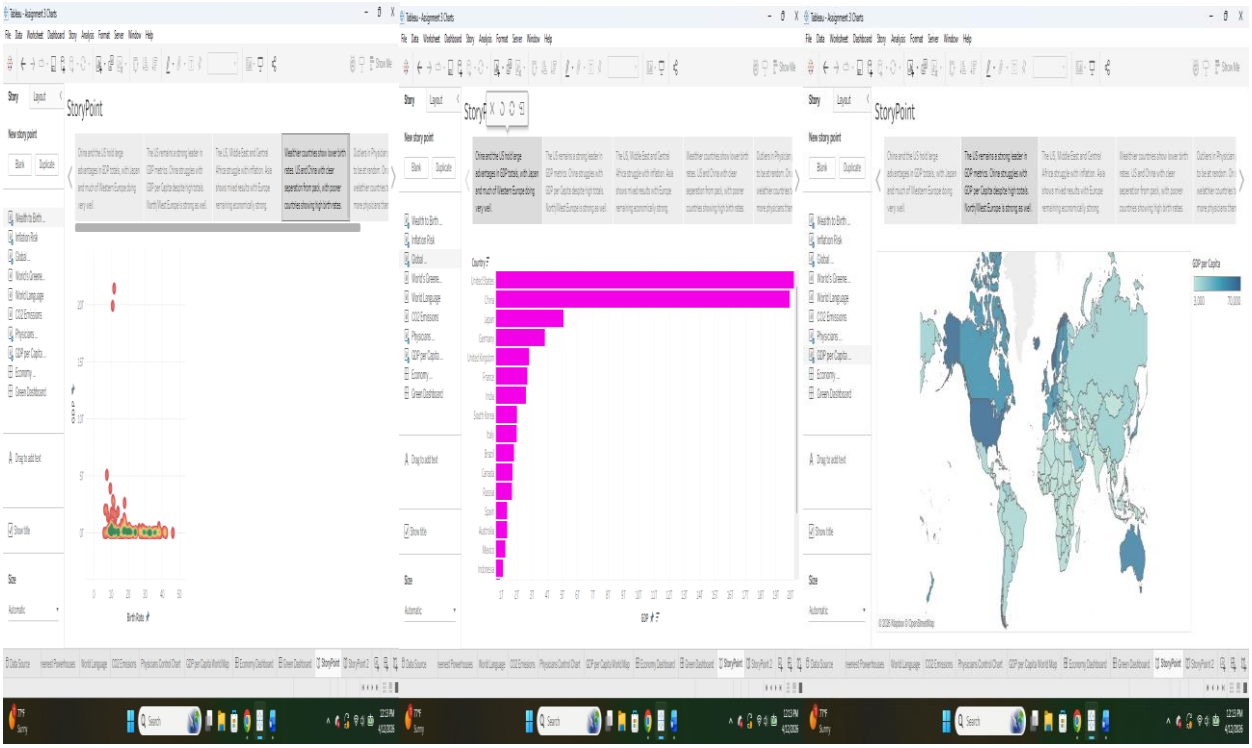
successful countries sport wealth and military power or conquest, but this dashboard dips into an often ignored and neglected division that determines the overall health and longevity of our existence.

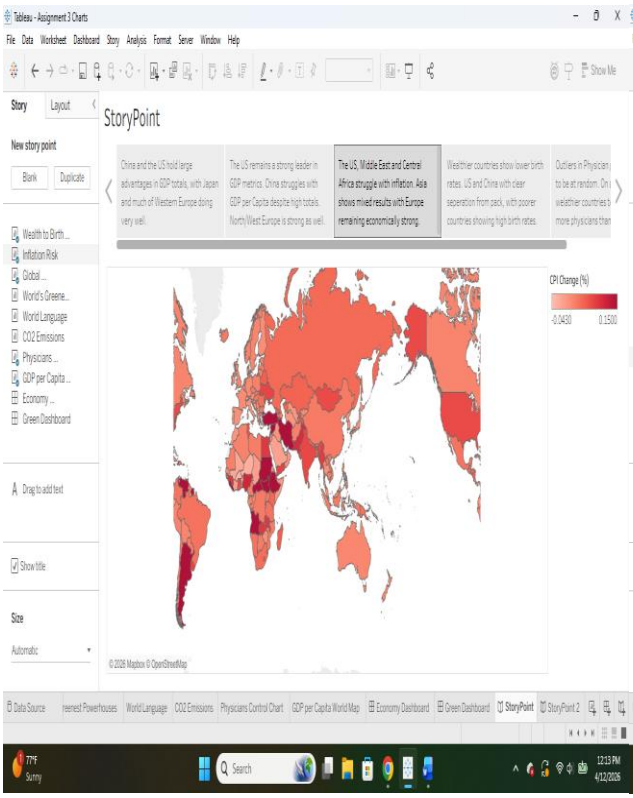
As shown in the upper-right bar chart, Asian and Asia-Pacific countries sit atop the countries preserving their forest and green homelands. Japan and South Korea hold large advantages over the field with well over 60% of their lands being forested. Brazil, Indonesia and Russia round out the top 5, with countries such as the UK, Australia and China surprisingly struggling to maintain forested lands. On the other end, Russia, the US and Germany surprisingly hold their own in terms of environmental preservations considering their knack for industrialization and factory production.

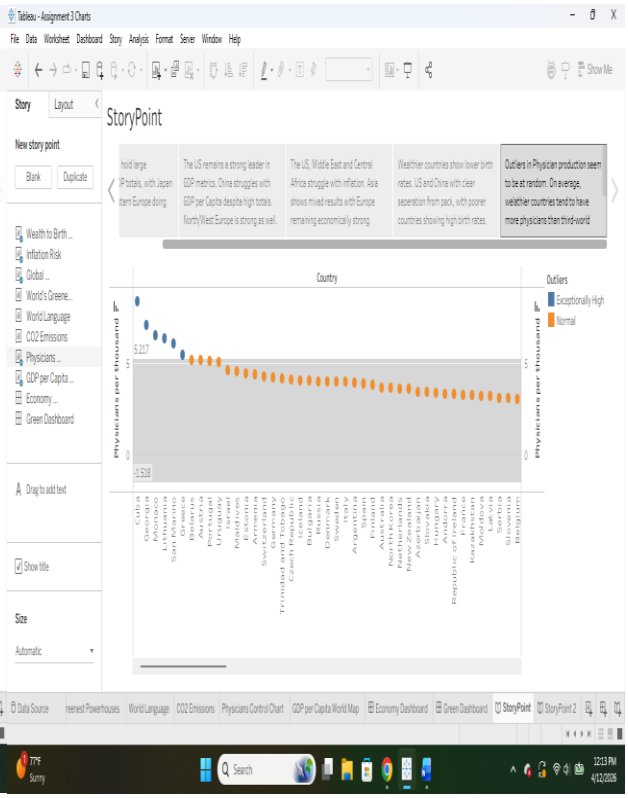
In the bottom pareto visual, you can see the overall carbon dioxide emissions from each country and the percentage of global emissions that pertain to them. Unsurprisingly, China and the US hold large amounts of total CO2 emission, accounting for over half of the global total. India, Russia and Japan round out that top 5, with many European nations being well down on the list limiting overall emission. When summarizing the results, it is clear that China consistently struggles to maintain environmental initiatives being the worst nation by far in each graphic. There are also many interesting nations with solid preserved forest but also high emission numbers such as the US, Russia and Japan.

6. Story Points

GDP Story Point







GDP Story

We began our international scope with identifying the most successful countries in the world according to their annual GDP totals. As shown in the pink bar chart, unsuprisingly the US and China dominated the field. Much of Europe covered the rankings, with occasional suprises showing up in India and Brazil. Identifying annual sum GDP was a solid start, but we knew there were underlying metrics to show who was truly prospering. This led to the creation of a world GDP per capita map (blue), displaying the most efficient countries economically.

To no surprise, the US and much of Western Europe continued to do very well. China on the other hand showed average totals, hinting at the overwhelmingly large population being the driving force for such a large annual GDP. India and Russia both displayed similar results. Sneaky prospering countries who missed the original top 20 GDP analysis included the Scandanavian nations in Northern Europe. Norway shockingly cracked the top 3 in word GDP per capita, with Finland, Sweden and Denmark all appearing in or near the top ten globally. With the GDP sources mapped out, we then decided to pivot towards a worldview of inflation risk through consumer price index change, or CPI change.

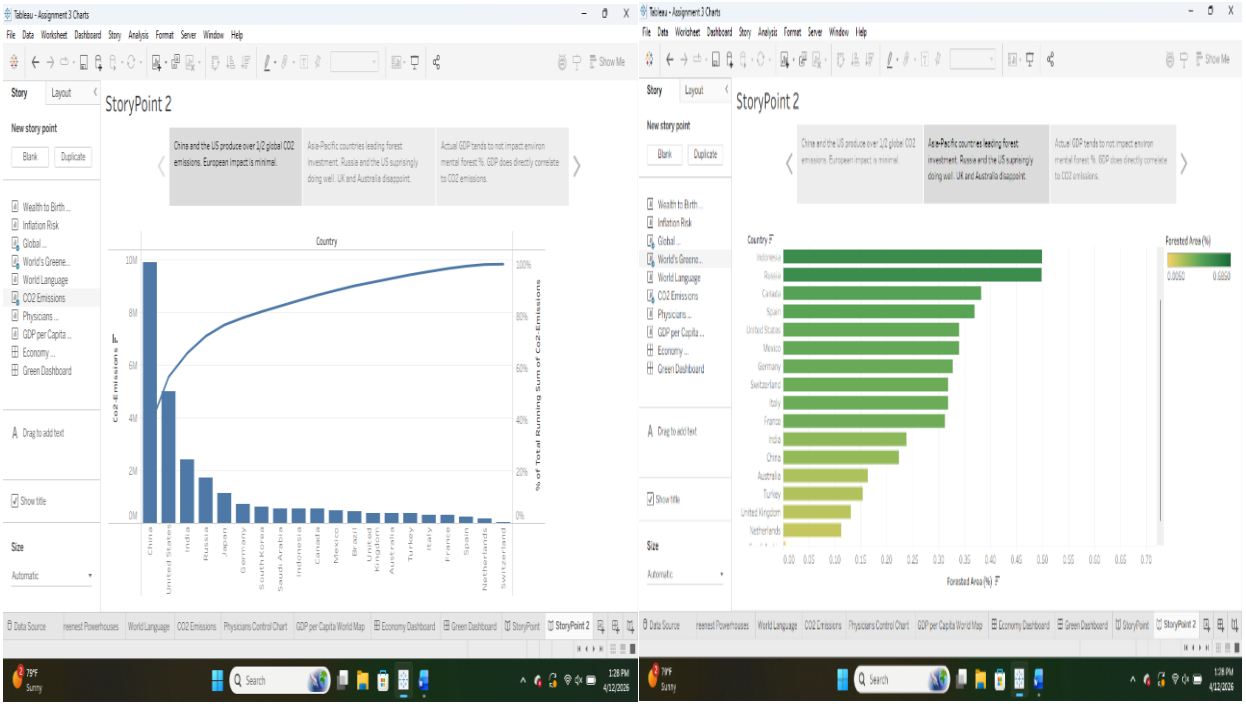
The inflation worldview map (red) showed not just regional struggle, but a worldwide inflation risk likely correlating with the effects of a global pandemic and potential US recession. With the US at nearly a full percentage change over the year, the entire world has shown to be affected with the global economy being so reliant upon the US dollar and imports/exports. Primary regions with extreme inflation included central Africa, South America, Southeast Asia and the Middle East. Much of Europe and China remain in between as well as Latin America, but these countries still show signs of potential inflation risk.

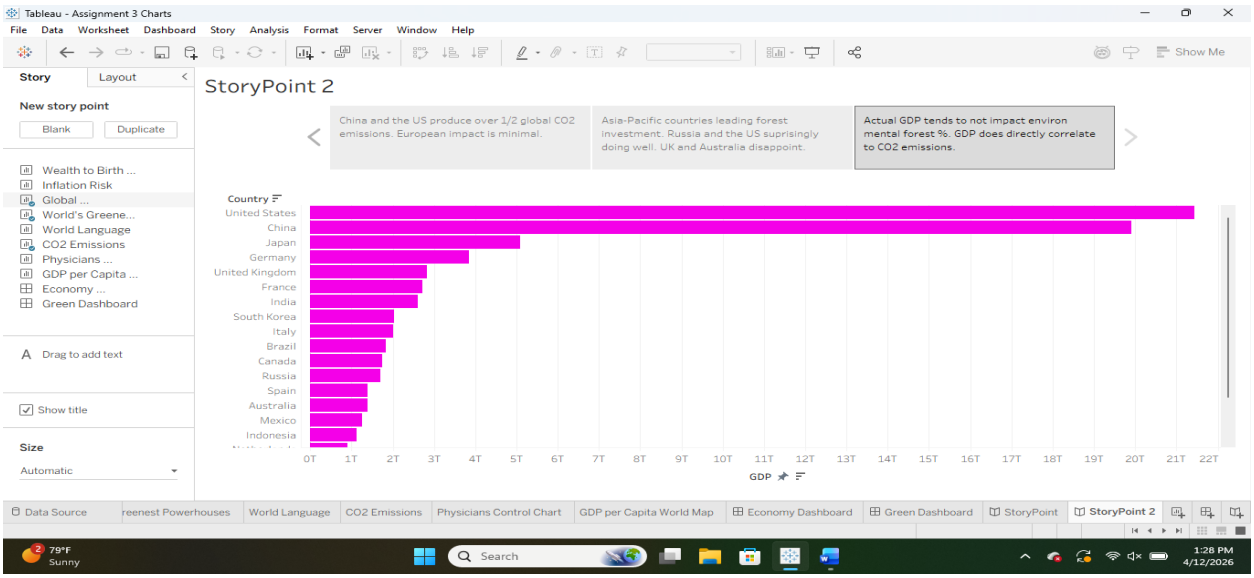
After reviewing the wealth disparity in the global economy, we decided to steer into how these affected other important demographics around the world. We did this by analyzing both birth ratios and physician presence. The density scatterplot shows the direct correlation between GDP and birth rates around the world. There is a clear inverse relationship, as wealthier countries tend to have lower birth rates with poorer countries having less control of their birth rates. Clear outliers furthering this point include the US, China and Japan. In our control chart

(orange and blue), we take a look at how well each country does at producing doctors, a premium field worldwide. Analyzing the chart, the outliers: Cuba, Monaco, Georgia, Lithuania, Greece and San Marino resulted in a fairly random result. None of these countries stand out in terms of GDP or inflation, and are all relatively average in regards to international state. However, within the control parameter there is a trend of first-world countries producing a more efficient rate of physicans compared to third-world countries. Much of Europe, and superpowers China, Russia and the US remain near the top of the control line while poorer nations in Africa and Southeast Asia remain near the bottom line.

Conclusively, the GDP track does its job to not only identify which countries are doing well, but also how these countries do in respect to their populations. Statistical analysis into birth rates and physician rates build upon the data to give a deeper look into how the wealth of these countries is impacting their day-to-day lives.

Environmental Story Point





Environmental Story

For this wave of information, we wanted to pivot away from a wealth perspective and move towards environmental concerns. We now live in a world with growing threats toward our planet's health and existance, and we wanted to know who was the main driver of these threats and what variables contributed to said threats.

Our first step was taking the intial GDP leaders in the world and analyzing how green they were on the surface. We did this by sorting the GDP leaders in descending order by amount of forested land in the country by percentage. The green visuals show just how green each nation is, with impressive dark green percentages at the top and disappointing yellowish results near the bottom. Upon analysis, Asia-pacific countries like Japan, South Korea and Indonesia led the pack with nearly two-thirds of their country being green. Suprises included highly industrial superpowers Russia and the US, with Russia being top 5 and the US being in the top 8 of the "World's Greenest Powerhouses" display. Disappointing returns included Australia at just over 16 percent dispite its flourishing wildlife, as well as the UK, China and the Netherlands.

Next, we wanted to dive a little deeper into the environmental cause. We did this by analyzing the CO2 emissions of each country worldwide and the percentage of emissions belonging to said countries. Using a pareto chart, we found that the US and China make up well over half of the world's CO2 emissions at over 56 percent. While not suprising these two countries were at the top, the mass total is concerning. Following China and the US were India, Russia and Japan, with the top 5 accounting for over 76 percent of CO2 emissions. The one silver lining was Europe. Aside from Germany, much of Europe had fairly low emission numbers and combined for under 12 percent combined.

Together, these story points give insight to the struggle that industrialization, specifically within the international powers, has presented the world with. High carbon dioxide emissions and deforestization worldwide has destroyed habitats and harmed the atmosphere. According to the data, it is up to the big players to address this issue and cut back on the harmful pollution that they can almost entirely be accredited with.

7. Effect of my Data Visualization

With my dataset pertaining to large amounts of data outside of your typical business sphere, my visualizations belong to a peculiar market. My analysis, specifically in the environmental sector, could be used by non-profit organizations centered around environmental preservation. Top environmental nonprofits interested in my visualizations could include 350org, The Climate Reality Project, Greenpeace, Sierra Club and many others.

My environmental dashboard and story point create a clear trend of excessive carbon dioxide emissions, especially amongst the international superpowers. Specifically, the emissions pareto chart does a good job of showing just how skewed the data is towards superpowers like the US and China. Together, according to my pareto chart, the US and China combine for well over half of the worlds total carbon dioxide emissions, with the next 18 wealthiest countries combining for only about 40 percent. Using this initiative, climate change driven non-profits like the ones mentioned above can convince federal governments and even international organizations of the issues the world is creating and the primary causes for them. Presenting these political bodies with further evaluation and research could work to potentially cut back on carbon dioxide emissions in these manufacturing giants to slow the destruction of our planet's health.

Exploring the other side of my visualizations, my economy analysis could also pivot to an economic consensus for major organizations such as the International Monetary Fund (IMF), World Bank or even World Trade Organization (WTO). Not only does my economy dashboard give you a snapshot of the top performing countries in the world by GDP and where they stand environmentally, but it also gives you insights into key metrics including inflation through consumer price index, annual GDP totals, GDP per Capita across the world and even niche categories like how wealth impacts birth rates as well as how physician output is related to these ratios. All this information put together gives a quick surface area detail for the state of the world economy from a diverse perspective involved everything from basic wealth production to population metrics and financial risk.

Conclusively, my visualizations are set up to be used on an international non-profit scale briefly overviewing the state of our world and how it is behaving both economically and environmentally.

8. Lessons Learned

Interestingly enough, the most difficult part of this project was actually finding my initial dataset to work with. On the surface, it is as simple as picking a dataset and running with it, but in reality, the entire project hinges upon the quality and perspectives your dataset offers. Not only did I have to find a dataset that interests me, but I also had to find one with enough data, diverse types of data and compatibility with Tableau and RStudio, which I personally ran into trouble with. My initial UFC dataset had almost strictly numerical data, and the MLB dataset I wanted to use could not load into Tableau due to reasons unknown. I was finally able to land on a simple csv file dataset in Kaggle, with a very diverse palette including a multitude of categorical and numerical data, as well as few null values and little to no corrupted data. The Global Country 2023 dataset not only had a mass amount of data and datatypes, but it also had a wide range of data in general ranging from GDP statistics to carbon dioxide emissions and birth rates. Fortunately, I was able to play around with a lot of different data and create my own calculated fields and graphs with relative ease.

Aside from the initial selection of data, the random forest machine learning algorithm gave me some trouble for one of my RStudio visualizations. The actual bar chart was simple, but running the model itself was tedious with a lot of code and error messages. Fortunately, I had done a lot of machine learning models in a previous data mining course, and was familiar with the interface of R. With some help from Gemini modifying code snippets, I was able to run the model and print the results to produce a reliable visualization for my random forest model.

The easiest part of this semester project was producing visualizations in Tableau. Everything from my initial charts to the dashboards and story points was relatively simple with Tableau's drag and drop interface, and I actually had a lot of fun playing with the data and visual appeals of my charts and graphs. Much of the work I have done with analytics up to this point involved

heavy coding and statistical breakdown, so the ease of simply playing around with data to create appealing visualizations was a breath of fresh air.

Considering the above confessions, I would say that I learned the actual dataset is more important than any of the analysis done afterwards. It is the foundation of your project and dictates the entirety of said project from step one all the way to the final submission. Choose Wisely!

9. My Experience

During my eight weeks working with Tableau and R, I learned a plethora of different ways to not only build visualizations but also mold them to my specific agendas. Tableau was by far the most valuable tool I learned from as I had already used R extensively in other courses. The drag and drop interface using the entire dataset from a side panel was a game changer and showed me why Tableau is one of the more popular visualization tools in the professional world.

I personally had a lot of fun building different types of plots with all sorts of different data. Once the charts were built, altering medians and colors was useful and really gave my plots some identity. The dashboards and story points were a unique way to bring all of it together into one conjunction, creating larger stories that entailed many different objectives into one body. My favorite trick with the dashboards was using the filters icon, which allowed you to look at one specific country or datapoint on each visualization in the dashboard; that was really neat and a lot of fun.

While I was already knowledgeable with the interface and workings of RStudio coming into the course, I did learn a lot about the additional tools available. Most of my previous work with R involved heavy coding centered around machine learning and regression models, and not much was made out of actual data visualization. Seeing how simple it was to build charts and graphs with different textures was eye-opening. Previously, many plots that came from my analysis were wonky and formatted in weird ways, especially when dealing with large decision trees and neural network plots. My work with R for this project showed me that not only can these plots be fitted, but they can also be formatted with different color schemes and chart types. The most

interesting feature I used was the plotly package that actually allowed my scatterplot to become interactive similar to Tableau. I had no idea R had that kind of capability and while Tableau is by far the better visualization tool, RStudio really has some useful stuff when it comes to visualizing deep learning models with plots.

Overall, throughout this course and project, I learned to take raw data and make it come alive. Professionally, this is a huge step as executives and potential clients do not want to see clusters of random data they do not understand. Being able to take massive amounts of data and analysis off of said data and bring it to life with visuals that keep individuals engaged is a game changer and opens a world of data communication not yet open to me before this project.

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